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HTML ABSTRACT

Thermophysical Property Measurements of Liquid Metals in Microgravity

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Abstract. The thermophysical properties of liquid metals can be studied by electromagnetic levitation, a containerless technique. It avoids contamination of the melt and gives access to the undercooled regime allowing accurate measurements of thermophysical properties sensitive to surface contamination, like surface tension. Microgravity helps to reduce the undesirable side effects of the levitation field, such as heating, stirring and shape deformation. During the SpaceLab missions IML-2 and MSL-1, density, surface tension, viscosity and electrical conductivity of selected metals and alloys was measured in the TEMPUS levitation facility using non-contact, videographic and inductive methods. The results obtained from experiments in microgravity prove the feasibility of the approach of containerless processing and give an indication of the high quality that can be obtained. However, for parametric studies of complex alloys, as used in industrial applications, a systematic research program is necessary. In the space station era, it will become possible to implement long-term research programs leading to, first, an optimisation of the experimental procedures and, second, to a comprehensive study of materials of scientific or technological interest. Under ESA's MAP program such application-oriented research projects are under way. This paper reviews the results available and discusses the objectives and progress achieved so far in the on-going preparatory programs.

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INTRODUCTION

The precise knowledge of thermophysical properties becomes increasingly important as progress is made in numerical simulations of complex processes. With the improvement of the models, their reliability is limited by the accuracy of the input parameters characterising the material under study. For high temperature melts, such as liquid metals, containerless methods are the best choice for their measurement. Among the different containerless techniques, electromagnetic levitation is especially suitable for the study of metallic melts (Herlach, 1993). It allows to obtain high temperatures, up to 2400 °C, to levitate bulk samples of a few grams, and to maintain the undercooled state for an extended period of time (up to hours).

In addition to containerless positioning and melting, also non-contact diagnostic tools have to be applied or developed. For instance, temperature is measured by the well known principle of radiation pyrometry (deWitt, 1989). In other cases, the necessary information is obtained by optical imaging or by inductive methods.

Some limitations exist for electromagnetic levitation when used on ground: the required high electromagnetic fields deform the shape of a molten sample, and induce turbulent currents inside the sample. In addition, the required fields are so strong that the samples must be cooled convectively using a high-purity inert gas, like He or Ar. All these drawbacks can be avoided if electromagnetic levitation is performed in microgravity. In such conditions, the positioning forces are reduced by at least a factor of 100, and the sample is essentially force-free during the cooling phase (The fields required for inductive heating are of course the same in 1g and in μg).

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MICROGRAVITY EXPERIMENTS

For containerless processing in microgravity, only restoring forces are required to counteract the effect of the random g-jitter and residual accelerations. The TEMPUS facility (Lohöfer, 1991) is an electromagnetic levitator for the microgravity environment. It uses a quadrupole field for positioning, which is established by feeding an electrical current through two sets of parallel coils of identical dimension in opposite direction. For such a configuration, the stabilizing field gradients are very strong, and the power absorption is very low. TEMPUS provides stable positioning against 10^{-2} g_0 , reducing the power absorbed in the sample by a factor of 100, as compared to the 1-g case. Heating is accomplished by a high-efficiency dipole field. The TEMPUS design allows to control heating and positioning of the sample independently, providing a quiescent sample during cooling. Temperature can be controlled over a wide range by adjusting the heating power, from approximately 400°C to 2400°C, depending on the material being processed.

The experiment unit of TEMPUS consists of an ultra high vacuum chamber which surrounds the levitation coils. Attached to the chamber through an axial and radial window are pyrometers and video cameras. 22 evaporation shields are mounted on a carrousel and are located in the optical path between pyrometers and sample and protect the pyrometers from contamination due to the evaporation from hot samples. They consist of either double mirrors or lenses. The samples are contained in sample holders, which are either wire cages or ceramic cups. 18 sample holders are loaded in the sample storage carrousel before flight. The video cameras are capable of either a high-speed mode or a high resolution mode. They provide axial (top) and radial (side) view of the sample. The experiment chamber is connected to the vacuum and gas systems, as well as to the power supply for the levitation circuit. TEMPUS has also its own Process and Data Management unit (PDM), which allows to run the experiments automatically from preprogrammed sequences, interactively by the astronauts through switches on its front panel, or remotely by near-real-time telecommanding from ground.

Surface Tension and Viscosity

To measure surface tension and viscosity the oscillating drop technique (Sauerland, 1992) is used. In microgravity, liquid samples perform oscillations around their spherical equilibrium shape. The radius of the sphere as a function of time reads: $R(t) = R_0 (1+\delta \cos(\omega t)\, e^{-\Gamma t})$. Simple formulae can be used to relate frequency ω and damping Γ of the oscillations of a sphere to surface tension γ and viscosity η, respectively. They are given by:

$$\omega^2 = \frac{32\pi}{3}\frac{\gamma}{M} \qquad (1)$$

and:

$$\Gamma = \frac{20\pi}{3}\frac{R_0\eta}{M} \qquad (2)$$

where M is the mass of the droplet and R_0 its radius. Under terrestrial conditions, the above relations are not valid; corrections have to be made for the external forces, namely gravity and electromagnetic field (Cummings 1991, and Bratz, 1995). The experiments are performed by applying short heating pulses during the cooling phase to excite oscillations. The sample oscillations are recorded with two video cameras, providing axial and radial view. The surface tension of $Co_{80}Pd_{20}$ measured during MSL-1 is shown in Figure 1. There are no terrestrial data available for this alloy. However, it is known from a comparative study (Egry, 1995) that terrestrial data yield too high values, unless they are corrected for the magnetic pressure effect mentioned above (Cummings, 1991).

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The same data set which is used for the determination of the surface tension can also be analysed with respect to viscosity, using eqn (2). The data collected during MSL-1 indicate that this method works remarkably well. For the alloy $Pd_{78}Cu_6Si_{16}$ two data sets were obtained, one at quiet conditions (low g-level), and one during a noisy period (high g-level). The results are shown in Figure 2. The scatter in the two datasets is distinctly different, however both can be fitted with the same Arrhenius-law: $\eta = \eta_0 \, exp(A/T)$. For comparison, also the result of Lee et al. (1991) from terrestrial measurements is shown and the agreement is excellent.

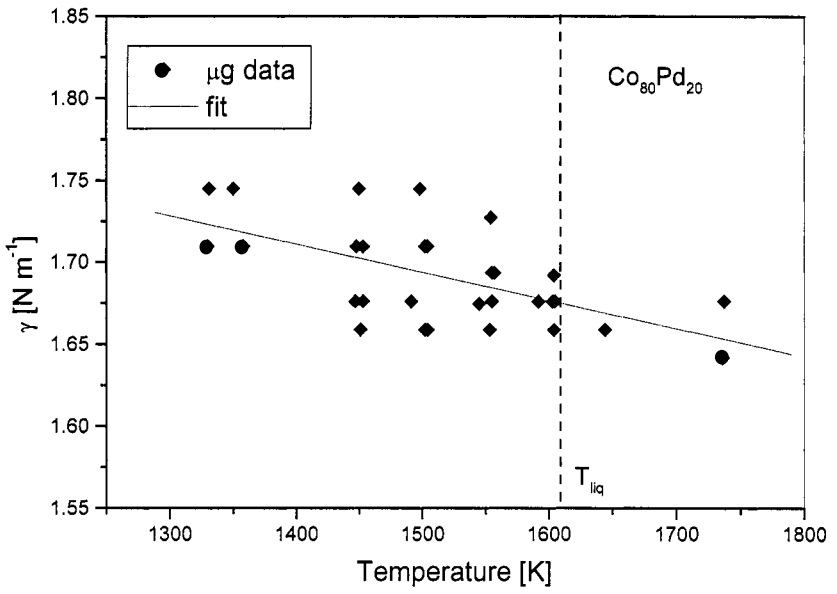

FIGURE 1. Surface tension of liquid and undercooled $Co_{80}Pd_{20}$. The dashed line indicates the liquidus temperature, T_{liq}.

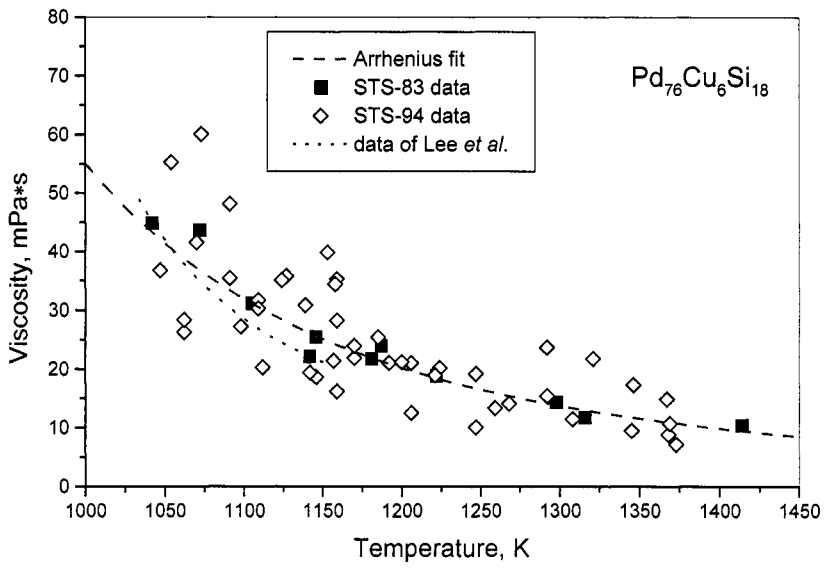

FIGURE 2. Viscsity of $Pd_{76}Cu_6Si_{18}$ measured during two different Spacelab missions, STS-83 and STS-94 and Arrhenius fit to the entire data set. For comparison, the result of Lee et al. (1991) is also shown.

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Thermal Expansion

A high resolution video camera is used to measure density and thermal expansion of levitated samples. It is equipped with telecentric optics which allows for absolute size measurements of moving objects. Pictures are taken from the side and the visible cross section of the sample is recorded. Assuming rotational symmetry, the volume V is calculated. Deviations from rotational symmetry due to non-symmetric surface oscillations (which are used for the surface tension measurements) have a sufficiently high frequency so that their influence can be suppressed by temporal averaging. Since the mass M of the sample is known and does not change, $\rho = M/V$ yields finally the density. This technique has been used to measure the thermal expansion of glass forming alloys in the undercooled regime. In order to observe any anomalies, a resolution of $\Delta V/V = 10^{-4}$ is required. This can be achieved using sub-pixel algorithms for edge detection, curve fitting of the shape, and statistical averaging (Damaschke, 1999).

During MSL-1 measurements were made on following samples: $Zr_{65}Cu_{17.5}Al_{7.5}Ni_{10}$, $Zr_{60}Ni_9Cu_{18}Al_{10}Co_3$, $Zr_{11}Ti_{34}Cu_{47}Ni_8$, $Zr_{57}Cu_{15.4}Ni_{12.6}Nb_5Al_{10}$, $Pd_{78}Cu_6Si_{16}$, and $Pd_{82}Si_{18}$. The result for $Zr_{60}Ni_9Cu_{18}Al_{10}Co_3$ is shown in figure 3. For the volumetric thermal expansion of the liquid and solid phases γ_s, and $\gamma_l = 1/V\ (\ V/\ T)$ the following values were fond: γ_s=2.11 10^{-4} [K^{-1}] and γ_l=4.73 10^{-5} [K^{-1}]. No terrestrial data exist for this alloy. However for the similar alloy $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10.0}Be_{22.5}$ a value for γ_l=5.32 10^{-5} [K^{-1}] was reported (Ohsaka, 1997).

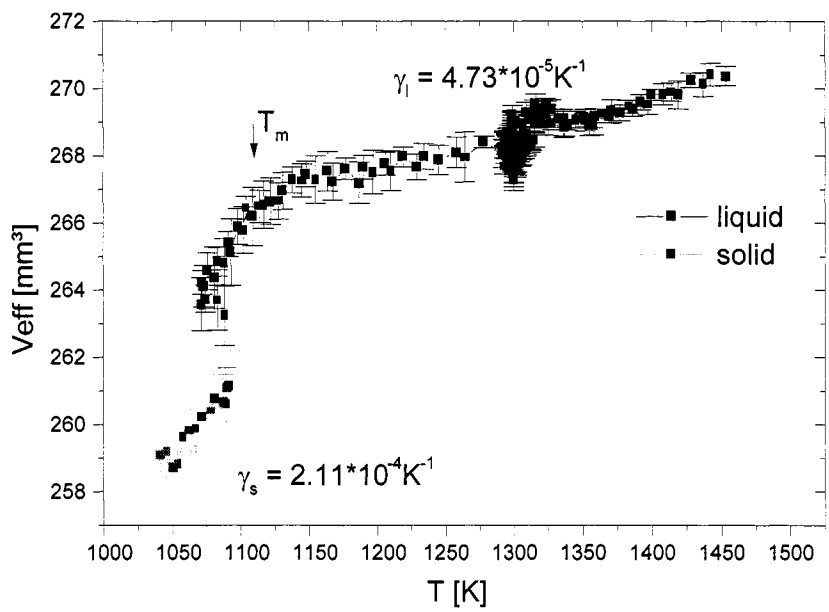

FIGURE 3. Density of liquid and solid $Zr_{60}Ni_9Cu_{18}Al_{10}Co_3$ -alloy.

Electrical Resistivity

Finally, it is also be possible to measure the electrical resistivity. This is done using a non-contact, inductive method, which is based on the fact that the levitated sample influences the impedance of the heating coils. Unfortunately, this is a very small effect and the signal-to-noise ratio is not very good. On the other hand, the sampling rate is quite high and the quality of the signal can be improved by statistical averaging. The impedance can be determined by measuring both, current I through the coil and voltage U across the coil simultaneously. One then obtains the amplitudes I_0 and U_0 of current and voltage, and the phase shift ϕ between them. Obviously, such measurements require that the current through the heating coil is not zero, $I_0 \neq 0$, and can therefore not be performed during free cooling. A small residual heating field is however sufficient. This was the case during the experiments on $Co_{80}Pd_{20}$ and $Zr_{60}Ni_9Cu_{18}Al_{10}Co_3$. The result for the latter alloy is shown in Figure 4. As can be

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seen, it shows a negative temperature coefficient of the resistivity in the liquid state, although its resistivity is below the Mooij limit (Mooij, 1973).

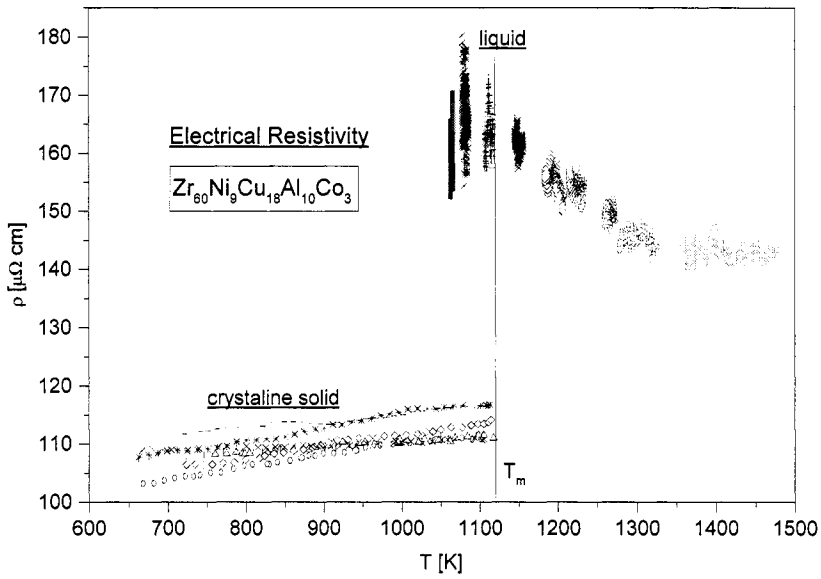

FIGURE 4. Electrical resistivity of $Zr_{60}Ni_9Cu_{18}Al_{10}Co_3$ alloy showing a negative temperature dependence in the liquid state.

SPACE STATION PLANNING

As the construction of the International Space Station progresses, also the payloads to be flown on board must be developed. Presently, there exist plans to provide containerless processing facilities for the ISS by acoustic, electrostatic and electromagnetic means. The Canadian Space Agency has put forward its SpaceDrums acoustic levitation facility, the Japanese Agency NASDA is developing an electrostatic levitator (ESL), while the European Space Agency ESA plans, jointly with the German Space Agency DLR, to build an electromagnetic levitation facility (EML) as part of their Materials Science Lab (MSL). For the measurement of thermophysical properties of liquid metals, the ESL and the EML are most suitable. In contrast to short-term missions, like Spacelab, these facilities must allow for long-term operation. This goal will be achieved by a modular design which will permit to exchange life-limited items and consumables. In addition, this approach also offers some flexibility to adapt the facility to the specific needs of particular users.

Apart from hardware development, also the utilization of these facilities is now in its preparatory phase. The permanent availability of levitators will lead to systematic research projects in the field of containerless processing. In order to promote the participation of industrial users, ESA has set up its Microgravity Application Promotion (MAP) program. One of these projects, ThermoLab (Fecht, 1999) is devoted to the measurement of thermophysical properties of liquid metals with industrial relevance. The project team consists of research labs from universities and government organisations, as well as of companies from the foundry and casting industry. The data obtained in microgravity will improve or complement existing datasets and will be used as input to numerical modelling software (Hepp, 2000).

CONCLUSIONS

The experiments with TEMPUS during the MSL-1 Spacelab mission have demonstrated that electromagnetic levitation in microgravity is a viable technology which provides insight into the properties of undercooled melts not

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obtainable by other methods. The results presented here prove the feasibility of the non-contact diagnostic methods developed for containerless processing. In some cases, the applicability of these methods remains restricted to experiments under microgravity, and will therefore rely on future space experiments. In other cases, results obtained in space can serve as benchmark tests and validation of terrestrial experiments combined with appropriate correction formulae. For the future, systematic studies of scientifically and technologically important alloys are proposed, using an Advanced Electromagnetic Levitation Facility on the International Space Station (ISS).

NOMENCLATURE

ω [1/s]	Angular frequency
γ [N/m]	Surface tension
M [kg]	Mass
Γ [1/s]	Relaxation time
R_0 [m]	Radius of sample
η [Pa s]	Viscosity
T [K]	Temperature
γ_l, γ_s [K^{-1}]	Volumetric thermal expansion of liquid and solid phase, respectively
ρ [Ω m]	Electrical resistivity

ACKNOWLEDGMENTS

Experimentation under microgravity is always a team effort and it is impossible to name everybody who contributed to our success. The results presented above would not have been achieved without the wunderful cooperation within and support from the entire TEMPUS team. Special thanks are due to the team from MUSC Cologne for their support during experiment preparation and experiment operations, to the engineers from Dornier Aerospace for building and operating the TEMPUS facility, and last but not least to the agencies, DLR and NASA, for providing us with a flight opportunity.

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